Caleb Geren

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Research interests

Zero-knowledge proof systems, succinct arguments

Education

2025 – Present **University of Utah** – Salt Lake City, UT
Ph.D. in Computer Science

Member of the Security & Privacy research cluster studying efficient privacy preserving proof systems (i.e. zero knowledge proofs, SNARKs). Generally working to make a wide variety of ZKPs more efficient in terms of time efficiency, space efficiency, or both.

2021 – 2025 **Lehigh University** – Bethlehem, PA
B.S. in Computer Science
B.A. in Mathematics

Publications

2026 **Succinct Arguments with (almost) Minimal Time and Space Overheads from Symmetric-Key Primitives**
Alexander R. Block, Caleb Geren, Pratik Soni
In Submission

2024 **Blockchain for Large Language Model Security and Safety: A Holistic Survey**
Caleb Geren, Amanda Board, Gaby G. Dagher, Tim Andersen, Jun Zhuang
ACM SIGKDD Explorations Newsletter, Volume 26, Issue 2

2023 **Scaling Zero-knowledge to Verifiable Databases**
Tal Derei, Benjamin Aulenbach, Victor Carolino, Caleb Geren, Michael Kaufman, Jonathan Klein, Rishad Islam Shanto, Henry F. Korth.
Proceedings of the 1st Workshop on Verifiable Database Systems

Research experience

Aug 2025 – Present **Kahlert School of Computing** *Building Time and Space Efficient zk-SNARKs*
Advisor: Pratik Soni

- Working to make zk-SNARKs efficient w.r.t. the Prover's space complexity, time complexity, and preserve their communication complexity.
- Developing time and space efficient arguments in zero-knowledge from error correcting codes.
- Helping to make zero-knowledge proof systems more compatible with floating point inputs.

Dec 2022 – May 2025 **Lehigh University Blockchain Lab** *Accelerating zk-SNARKs on GPUs*
Mentor: Henry F. Korth

- An attempt to parallelize the zk-SNARK proving system Plonk in order to scale zero-knowledge systems generally.
- Preliminary results from existing Plonk implementations gave way to promising avenues towards an implementation of the Plonk prover which exploits a multi-streaming technique on GPUs. After exploration, the multi-streaming technique proved infeasible in light of certain computational bottlenecks in the underlying Plonk computation.

May 2024 - July 2024 **Boise State University Blockchain REU** *Blockchain for LLMs: A Survey*
Mentor: Gaby Dagher

- A thorough exploration into the complementary nature of blockchain in the realm of large language model safety and security was conducted.
- Resulted in a lead author KDD '24 publication, with much of the drafting, outline, and research questions developed independently.

Honors and scholarships

2025 Honorable Mention - Graduate Research Fellowship Program (GRFP; GRFP site)
Honorable mention in the extremely competitive 2024-2025 GRFP cycle where my project proposol focused on novel approaches to accelerating zk-SNARKs on modern architectures.

2024 Boise State University Blockchain REU (BREU; BREU site)
REU fellowship which provided me with the opportunity to develop blockchain research concerning large language model applications.

2023 DAAD Rise Scholarship Recipient (DAAD; RISE site)
International scholarship which pairs undergraduate students from North America, Great Britain, and Ireland with faculty at German universities for summer research.

Teaching experience

Fall 2023 Fall 2024 **Teaching assistant, CSE 242: Blockchain Systems (Lehigh University)**
Delivered a lecture on the details of the Plonk proving system (see Talks and Posters), as well as held office hours and graded assignments on a weekly basis.

Spring 2024 **Teaching assistant, CSE 340: Algorithms (Lehigh University)**
Held twice weekly office hours reviewing homework assignments, course content, and grading assignments.

Talks and Posters

August 2024 **Poster: Integrating Blockchain with LLMs: Towards a Secure and Safe Technology**
Poster given at the Idaho Conference for Undergraduate Research (ICUR).

November 2023 **Lecture: An In-Depth Look at the Plonk Zero-knowledge Proving System: Plonk By Hand**
Lecture delivered in the Blockchain Systems course CSE 242 covering the details of the zk-SNARK construction Plonk.

August 2023 **Poster: Blockchain Systems and Applications Research**
Presentation given at Lehigh Summer Research Internship.

April 2023 **Poster: Scaling Zero-Knowledge Proof Generation for Large Blockchain Applications**
Presented to Lehigh University's internal I-DISC conference.

Other interests

Besides my passion for cryptography, I also love to rock climb, snowboard, and canyoneer.

Updated: October 26, 2026